

模式识别与机器学习

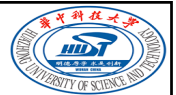
授课教师：杨卫东 胡静

L5-1



1

第六讲 非线性变换 (*Nonlinear Transformation*)



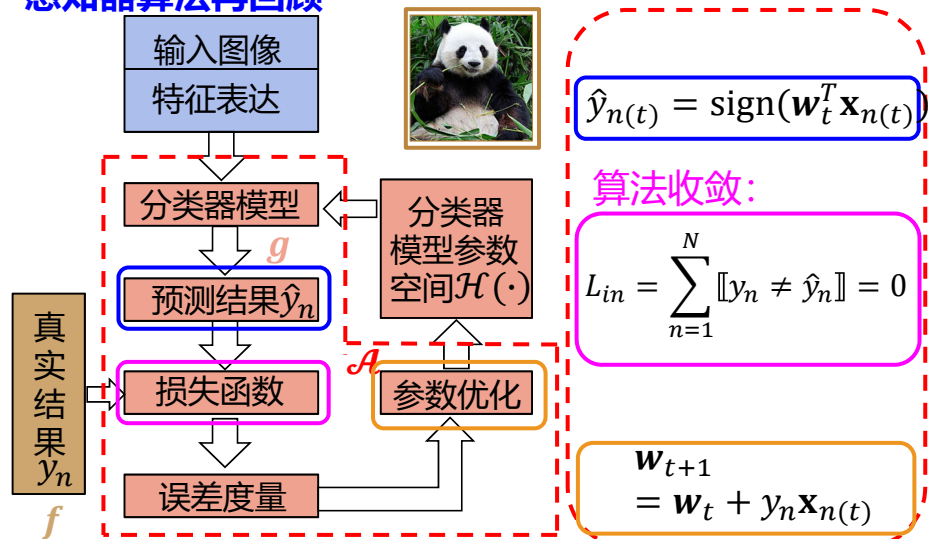
6.1 线性不可分问题 (*Nonlinear Data Problem*)

6.2 非线性变换 (*Nonlinear Transform*)

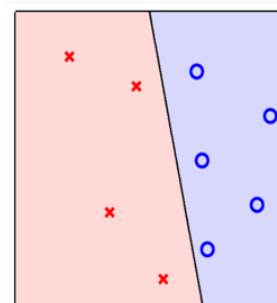
6.3 知识拓展 (*Knowledge Extension*)

6.1 线性不可分问题

感知器算法再回顾

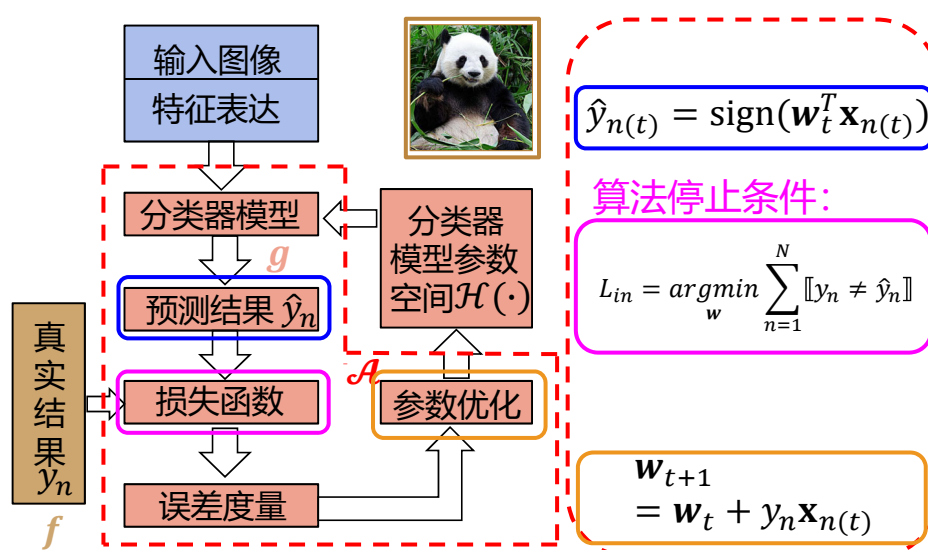


线性可分

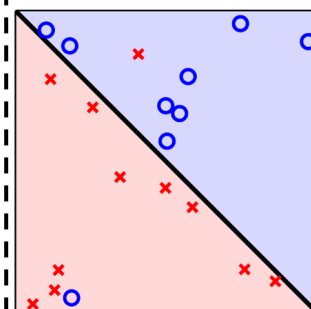


- 设置初始分类面 (权重) $\mathbf{w}_0$
- 如果有样本分错, 就修正权重

6.1 线性不可分问题

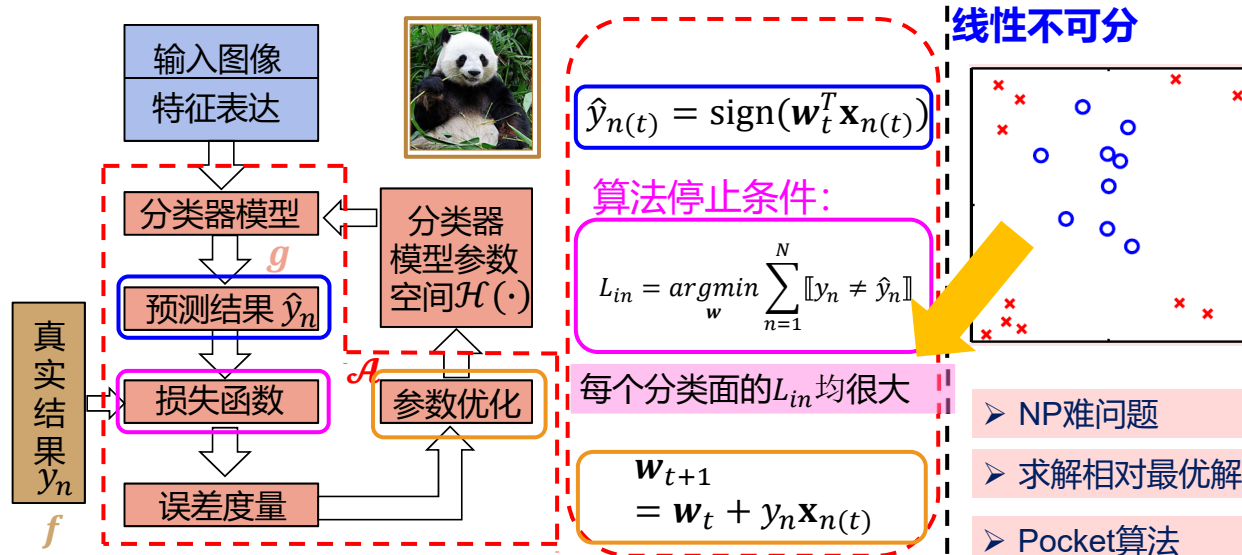
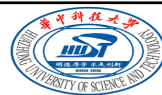


线性不可分



- NP难问题
- 求解相对最优解
- Pocket算法

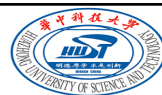
6.1 线性不可分问题



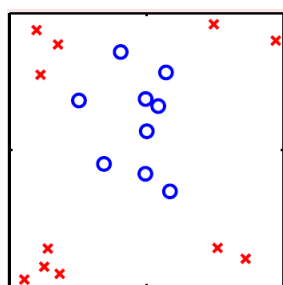
人工智能与自动化学院

5

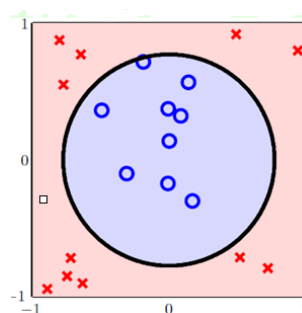
6.1 线性不可分问题



如何突破线性分类限制



线性不可分



圆圈可分

$$h_{sep}(\mathbf{x}) = \text{sign}(-x_1^2 - x_2^2 + 0.6)$$

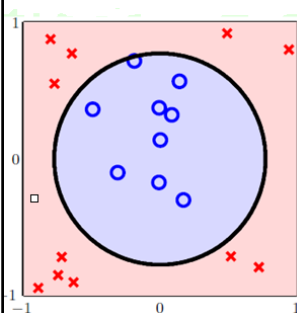
人工智能与自动化学院

6

6.1 线性不可分问题

圆圈可分与线性可分

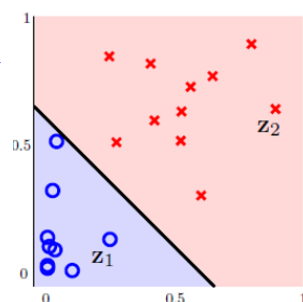
$$\begin{aligned} h_{sep}(\mathbf{x}) &= \text{sign}(0.6 \cdot 1 + (-1) \cdot x_1^2 + (-1) \cdot x_2^2) \\ &= \text{sign}(\tilde{w}_0 z_0 + \tilde{w}_1 z_1 + \tilde{w}_2 z_2) = \text{sign}(\tilde{\mathbf{w}}^T \mathbf{z}) \end{aligned}$$



$\{(\mathbf{x}_n, y_n)\}$ 圆圈可分 $\Rightarrow \{(\mathbf{z}_n, y_n)\}$ 线性可分

$$\mathbf{x} \in \mathcal{X} \xrightarrow{\Phi} \mathbf{z} \in \mathcal{Z}$$

Φ : 非线性特征变换



6.1 线性不可分问题

利用二次多项式的一般表达将样本 $\mathbf{x}$ 从 $\mathcal{X}$ 空间变换到 $\mathcal{Z}$ 空间

$$\Phi_2(\mathbf{x}) = (1, x_1, \dots, x_d, x_1^2, x_1 x_2, \dots, x_d^2)^T$$

如果样本 $\mathbf{x}$ 是 2 维特征, 则: $\Phi_2(\mathbf{x}) = (1, x_1, x_2, x_1^2, x_1 x_2, x_2^2)^T$

样本 $\mathbf{x}$ 从原来的 d 维特征空间变换到多少维特征空间?

$$\tilde{d} = 1 + d + d + C_d^2 = 1 + d + d + \binom{d}{2}$$

不放回的组合问题

$$= 1 + d + d + \frac{d(d-1)}{2} = 1 + \frac{d(d+3)}{2} = \frac{(d+2)(d+1)}{2}$$

6.1 线性不可分问题

利用二次多项式的一般表达将样本 $\mathbf{x}$ 从 $\mathcal{X}$ 空间变换到 $\mathcal{Z}$ 空间

$$\Phi_2(\mathbf{x}) = (1, x_1, \dots, x_d, x_1^2, x_1x_2, \dots, x_d^2)^T$$

如果样本 $\mathbf{x}$ 是 2 维特征, 则: $\Phi_2(\mathbf{x}) = (1, x_1, x_2, x_1^2, x_1x_2, x_2^2)^T$

样本 $\mathbf{x}$ 从原来的 d 维特征空间变换到多少维特征空间?

$$\tilde{d} = 1 + d + d + C_d^2 = 1 + d + d + \binom{d}{2}$$

放回的组合问题

$$= 1 + d + d + \frac{d(d-1)}{2} = 1 + \frac{d(d+3)}{2} = \frac{(d+2)(d+1)}{2} = \binom{2+d}{2}$$

6.1 线性不可分问题

利用 Q 次多项式的一般表达将样本 $\mathbf{x}$ 从 $\mathcal{X}$ 空间变换到 $\mathcal{Z}$ 空间

$$\Phi_Q(\mathbf{x}) = (1, \underbrace{x_1, \dots, x_d}_{\text{一次项}}, \underbrace{x_1^2, x_1x_2, \dots, x_d^2}_{\text{二次项}}, \dots, \underbrace{x_1^Q, x_1^{Q-1}x_2, \dots, x_d^Q}_{\text{Q次项}})^T$$

样本 $\mathbf{x}$ 从原来的 d 维特征空间变换到多少维特征空间?

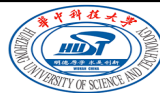
$$\tilde{d} = C_{Q+d}^Q = \binom{Q+d}{Q}$$

放回的组合问题

非线性变换
使特征被升
到高维空间

$$= \frac{(Q+d)!}{Q! \cdot d!} = \frac{(Q+d-1)(Q+d-2) \cdots (Q+1)}{d!} \Rightarrow Q^d$$

第六讲 非线性变换 (Nonlinear Transformation)

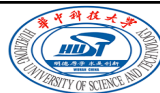


6.1 线性不可分问题 (Nonlinear Data Problem)

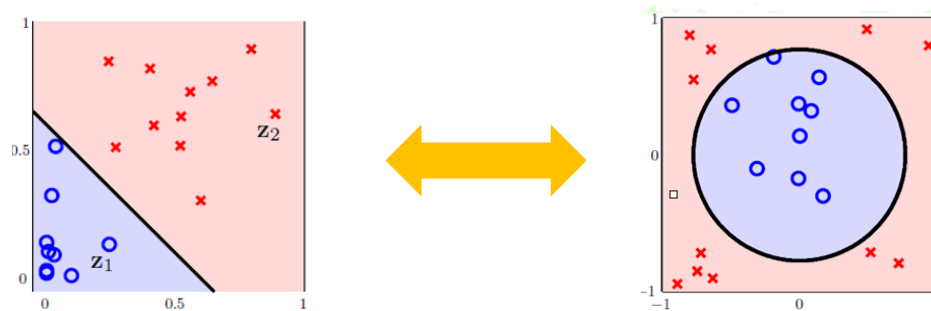
6.2 非线性变换 (Nonlinear Transform)

6.3 知识拓展 (Knowledge Extension)

6.2 非线性变换



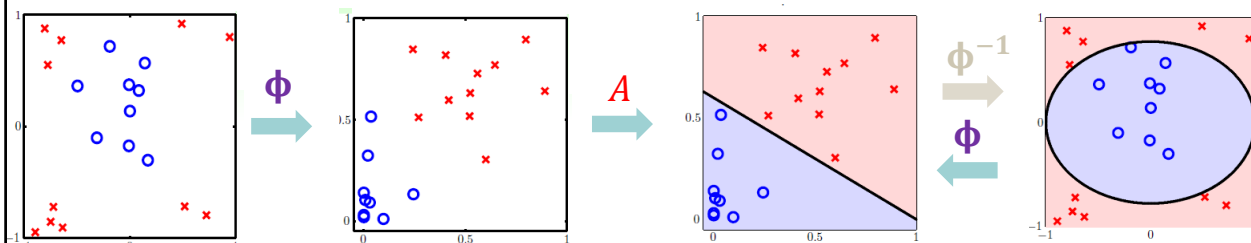
非线性变换的目的



通过非线性变换 Φ_Q 使得训练样本集 $\{(z_n = \Phi_Q(x_n), y_n)\}$ 在 z 空间找到好的分类面

6.2 非线性变换

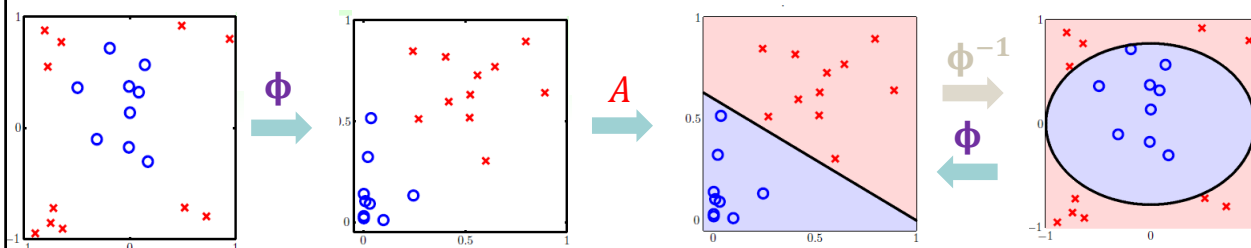
非线性变换步骤



- ① 利用非线性变换 ϕ 将原始训练样本集 $\{(\mathbf{x}_n, y_n)\}$ 变换到 Z 空间 $\{(\mathbf{z}_n = \phi(\mathbf{x}_n), y_n)\}$;
- ② 在数据集 $\{(\mathbf{z}_n, y_n)\}$ 上选择合适的线性分类算法 A , 得到最佳解 $\tilde{\mathbf{w}}^*$
- ③ 返回分类结果: $g(\mathbf{x}) = \text{sign}(\tilde{\mathbf{w}}^{*T} \mathbf{x})$

6.2 非线性变换

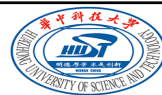
非线性模型 $\rightarrow$ 非线性变换 ϕ + 线性模型



线性模型不局限于二元分类;

通过非线性变换, 可以方便地实现: 二次PLA、三次PLA、更高次数多项式的PLA
二次回归、三次回归、更高次数回归...

6.2 非线性变换



特征提取



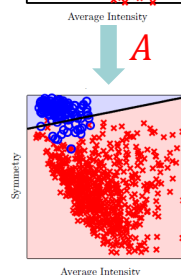
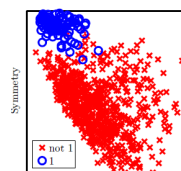
特征变换 ϕ

非线性变换并不一定是多项式变换

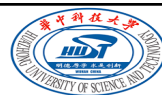
图像原始像素值
raw (pixels)

领域知识
domain knowledge

具体特征
concrete (intensity, symmetry)



第六讲 非线性变换 (Nonlinear Transformation)

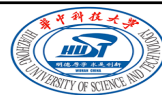


6.1 线性不可分问题 (Nonlinear Data Problem)

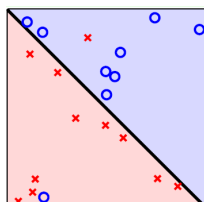
6.2 非线性变换 (Nonlinear Transform)

6.3 知识拓展 (Knowledge Extension)

6.3 知识拓展



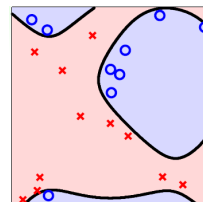
模型泛化能力讨论(Generalization Issue)



Φ_1 (original $\mathbf{x}$)

$$L_{in} \neq 0$$

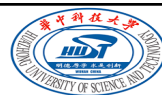
你认为哪个分类面更好?



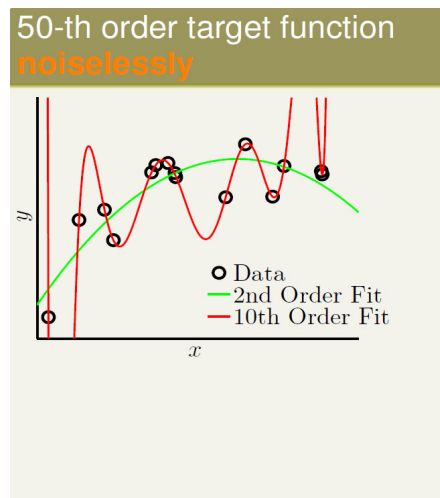
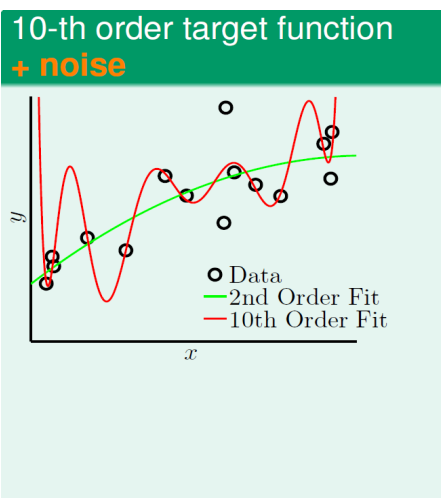
Φ_4

$$L_{in} = 0$$

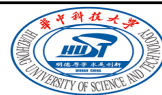
6.3 知识拓展



模型泛化能力讨论(Generalization Issue)



6.3 知识拓展



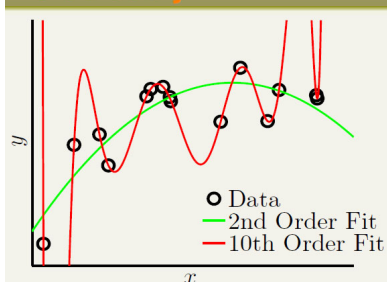
模型泛化能力讨论(Generalization Issue)

10-th order target function
+ noise



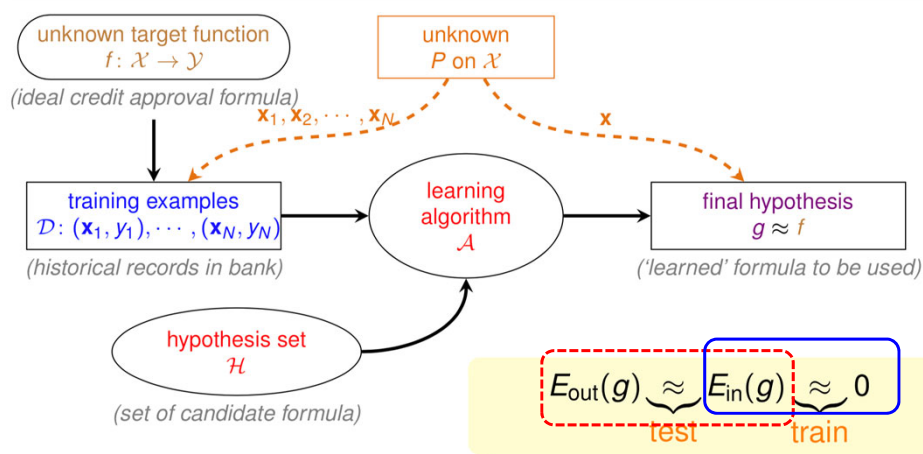
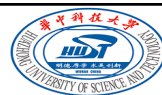
	$g_2 \in \mathcal{H}_2$	$g_{10} \in \mathcal{H}_{10}$
L_{in}	0.050	0.034
L_{out}	0.127	9.00

50-th order target function
noiselessly



	$g_2 \in \mathcal{H}_2$	$g_{10} \in \mathcal{H}_{10}$
L_{in}	0.029	0.00001
L_{out}	0.120	7680

机器学习的过程



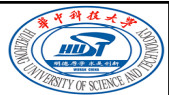
机器学习的过程



- ❖ **期望误差**: $E_{\text{out}}(g)$, 学到的假设 g 在除了训练样本外的其他所有样本(out-of-sample)上的损失, 也称泛化误差
- ❖ **经验误差**: $E_{\text{in}}(g)$, 学到的假设 g 在训练样本(in-of-sample)上的损失
- ❖ 可学习条件, 样本足够多
 - (1) train过程, 希望**经验误差**尽可能小
 - (2) test过程, 希望在真实环境中的**期望误差**与经验误差一致

21

6.3 知识拓展

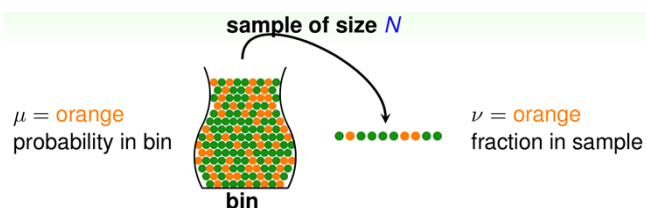


Hoeffding's Inequality (霍夫丁不等式, 数学家)

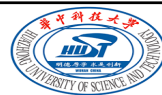
- in big sample (N large), ν is probably close to μ (within ϵ)

$$\mathbb{P}[|\nu - \mu| > \epsilon] \leq 2 \exp(-2\epsilon^2 N)$$

http://blog.csdn.net/qq_34993631



6.3 知识拓展



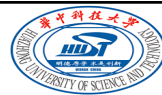
模型泛化能力讨论(Generalization Issue)

Vapnik-Chervonenkis (VC) Bound

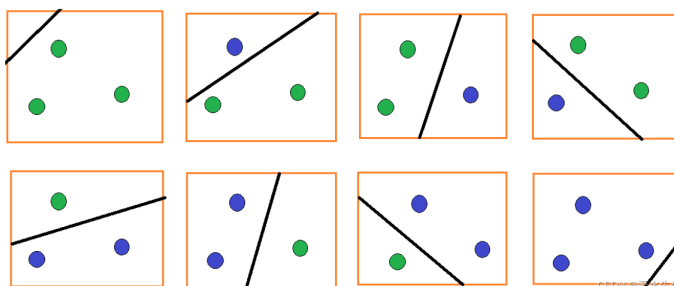
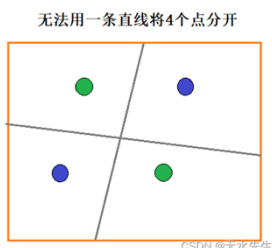
For any $g = \mathcal{A}(\mathcal{D}) \in \mathcal{H}$ and 'statistical' large $\mathcal{D}$, for $N \geq 2, d_{VC} \geq 2$

$$\mathbb{P}_{\mathcal{D}} \left[\underbrace{|E_{in}(g) - E_{out}(g)|}_{\text{BAD}} > \epsilon \right] \leq \underbrace{4(2N)^{d_{VC}} \exp\left(-\frac{1}{8}\epsilon^2 N\right)}_{\delta}$$

VC维



- ❖ 数据样本的属性维度d
- ❖ H个样本能够被函数集S中的函数按所有可能的 2^H 分开，正确二分类
- ❖ H个样本可以任意给出，无条件限制



d=2, H=3

6.3 知识拓展



Vapnik-Chervonenkis (VC) Bound

For any $g = \mathcal{A}(\mathcal{D}) \in \mathcal{H}$ and 'statistical' large $\mathcal{D}$, for $N \geq 2, d_{VC} \geq 2$

$$\mathbb{P}_{\mathcal{D}} \left[\underbrace{|E_{in}(g) - E_{out}(g)| > \epsilon}_{\text{BAD}} \right] \leq \underbrace{4(2N)^{d_{VC}} \exp\left(-\frac{1}{8}\epsilon^2 N\right)}_{\delta}$$

..., with probability $\geq 1 - \delta$, **GOOD**: $|E_{in}(g) - E_{out}(g)| \leq \epsilon$

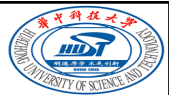
$$\text{set } \delta = 4(2N)^{d_{VC}} \exp\left(-\frac{1}{8}\epsilon^2 N\right)$$

$$\frac{\delta}{4(2N)^{d_{VC}}} = \exp\left(-\frac{1}{8}\epsilon^2 N\right)$$

$$\ln\left(\frac{4(2N)^{d_{VC}}}{\delta}\right) = \frac{1}{8}\epsilon^2 N$$

$$\sqrt{\frac{8}{N} \ln\left(\frac{4(2N)^{d_{VC}}}{\delta}\right)} = \epsilon$$

6.3 知识拓展



Vapnik-Chervonenkis (VC) Bound

For any $g = \mathcal{A}(\mathcal{D}) \in \mathcal{H}$ and 'statistical' large $\mathcal{D}$, for $N \geq 2, d_{VC} \geq 2$

$$\mathbb{P}_{\mathcal{D}} \left[\underbrace{|E_{in}(g) - E_{out}(g)| > \epsilon}_{\text{BAD}} \right] \leq \underbrace{4(2N)^{d_{VC}} \exp\left(-\frac{1}{8}\epsilon^2 N\right)}_{\delta}$$

..., with probability $\geq 1 - \delta$, **GOOD!**

$$\text{gen. error } |E_{in}(g) - E_{out}(g)| \leq \sqrt{\frac{8}{N} \ln\left(\frac{4(2N)^{d_{VC}}}{\delta}\right)}$$

$$E_{in}(g) - \sqrt{\frac{8}{N} \ln\left(\frac{4(2N)^{d_{VC}}}{\delta}\right)} \leq E_{out}(g) \leq E_{in}(g) + \sqrt{\frac{8}{N} \ln\left(\frac{4(2N)^{d_{VC}}}{\delta}\right)}$$

6.3 知识拓展



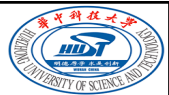
模型复杂度

with a high probability,

$$E_{\text{out}}(g) \leq E_{\text{in}}(g) + \underbrace{\sqrt{\frac{8}{N} \ln \left(\frac{4(2N)^{d_{\text{VC}}}}{\delta} \right)}}_{\Omega(N, \mathcal{H}, \delta)}$$

$\underbrace{\sqrt{\dots}}_{\Omega(N, \mathcal{H}, \delta)}$: penalty for **model complexity**

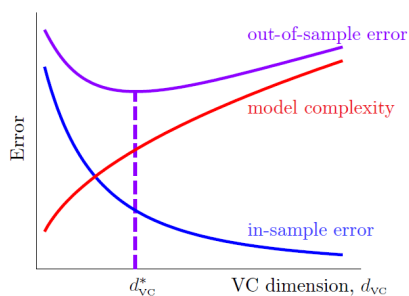
6.3 知识拓展



模型复杂度

with a high probability,

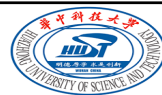
$$E_{\text{out}}(g) \leq E_{\text{in}}(g) + \underbrace{\sqrt{\frac{8}{N} \ln \left(\frac{4(2N)^{d_{\text{VC}}}}{\delta} \right)}}_{\Omega(N, \mathcal{H}, \delta)}$$



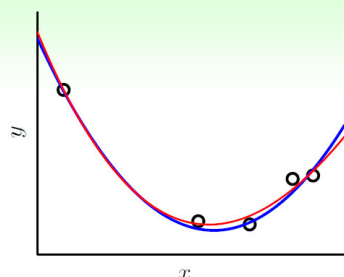
- $d_{\text{VC}} \uparrow$: $E_{\text{in}} \downarrow$ but $\Omega \uparrow$
- $d_{\text{VC}} \downarrow$: $\Omega \downarrow$ but $E_{\text{in}} \uparrow$
- best d_{VC}^* in the middle

powerful $\mathcal{H}$ not always good!

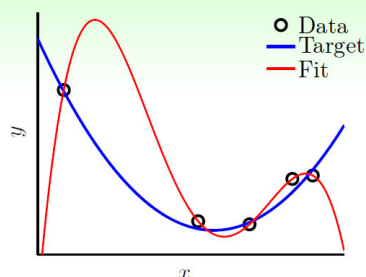
6.3 知识拓展



泛化性能不好与过拟合(Bad Generalization and Overfitting)

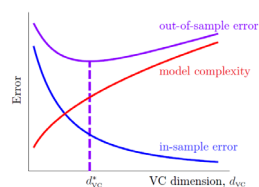


'good fit'

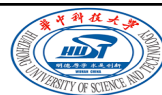


overfit

bad generalization: low E_{in} , high E_{out} ;
overfitting: lower E_{in} , higher E_{out}

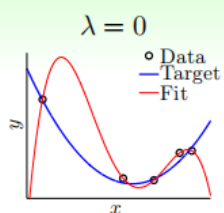


6.3 知识拓展

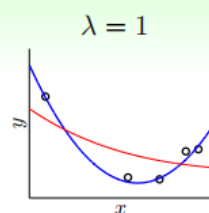
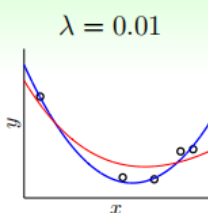
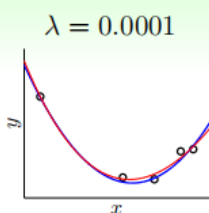


过拟合与正则化(Overfitting and Regularization)

$$\min_{\mathbf{w}} E_{\text{aug}}(\mathbf{w}) = E_{\text{in}}(\mathbf{w}) + \frac{\lambda}{N} \mathbf{w}^T \mathbf{w}$$

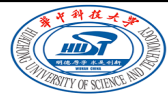


overfitting

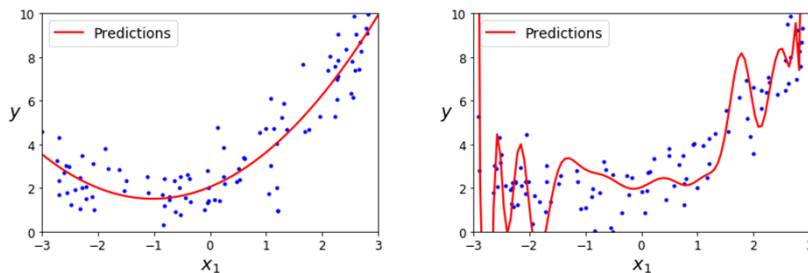


underfitting

6.3 知识拓展

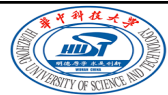


❖ 正则化(Regularization)



- 数据去噪
- 降维，减少特征，留取最重要的特征
- 惩罚不重要的特征的权重

第六讲 非线性变换 (Nonlinear Transformation)



6.1 线性不可分问题 (Nonlinear Data Problem)

通过多项式变换后的数据集符合线性模型特点

6.2 非线性变换

利用 $Z = \Phi(X)$ 变换后，可以方便地使用线性模型处理

6.3 知识拓展

VC Bound、模型复杂度、泛化能力、过拟合、正则化